

STE AGATHE DES MONTS, QC, Canada

WMO: 727200

Lat: 46.050N

Lon: 74.280W

Elev: 1295

StdP: 14.02

Time Zone: -5.00 (NAE)

Period: 82-92

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.5	-14.0	-27.3	1.3	-17.9	-23.0	1.6	-13.0	24.7	10.2	21.6	13.7	4.6	300	0.608

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	81.5	68.2	78.8	66.8	76.2	65.1	70.8	77.7	69.1	75.3	67.4	73.4	8.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.7	110.7	74.1	66.9	104.0	72.3	65.1	97.4	70.7	35.5	77.5	34.0	75.0	32.6	73.5	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	17.6	15.5	-26.2	86.0	2.6	2.9	-28.1	88.1	-29.6	89.8	-31.1	91.4	-32.9	93.5
			-26.6	73.4	2.6	2.2	-28.4	75.0	-29.9	76.2	-31.4	77.5	-33.2	79.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.4	9.4	13.6	22.5	38.3	51.3	58.9	64.4	61.7	53.0	42.5	29.5	14.2
	DBStd	21.44	12.95	11.32	12.74	9.62	8.15	7.00	5.78	6.50	7.44	7.63	9.33	12.75
	HDD50	5630	1259	1020	853	371	83	9	0	4	50	255	615	1111
	HDD65	9850	1724	1440	1318	802	430	210	81	143	366	696	1064	1576
	CDD50	1394	0	0	0	19	124	277	445	365	139	24	1	0
	CDD65	140	0	0	0	0	5	28	61	40	6	0	0	0
	CDH74	1059	0	0	0	12	66	245	469	236	31	0	0	0
	CDH80	149	0	0	0	0	6	44	84	14	1	0	0	0
Wind	WSAvg	6.4	6.9	6.4	7.2	6.9	6.2	6.2	5.4	5.0	5.6	6.3	7.3	7.0
Precipitation	PrecAvg	48.7	3.9	3.4	3.7	3.4	3.7	4.4	4.0	4.4	4.1	3.9	4.8	4.8
	PrecMax	62.4	7.3	8.9	5.4	5.5	8.0	8.9	8.8	9.2	6.6	6.2	9.1	7.7
	PrecMin	41.8	1.5	0.8	1.0	1.1	1.1	1.6	1.5	1.0	1.5	2.2	3.1	2.3
	PrecStd	5.2	1.5	2.0	1.1	1.2	1.8	1.7	1.7	1.9	1.3	1.1	1.4	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.5	42.9	57.8	75.5	80.6	84.6	86.1	81.7	78.3	67.8	57.0	48.7
		MCWB	38.7	37.2	48.1	60.1	61.1	67.6	71.3	69.7	67.1	59.9	52.1	47.4
	2%	DB	34.7	38.2	48.4	66.0	75.0	80.5	82.2	79.2	73.0	63.4	52.1	39.4
		MCWB	33.1	34.8	42.9	53.8	60.3	66.5	69.7	67.7	64.9	57.1	50.0	38.2
	5%	DB	32.1	34.4	43.4	59.6	70.9	76.5	79.2	76.8	69.8	59.0	47.8	34.5
		MCWB	31.2	32.1	38.1	48.4	57.8	64.2	67.7	66.6	63.7	53.8	45.9	32.8
	10%	DB	28.8	31.7	38.7	54.0	66.8	72.6	76.1	74.0	66.3	55.2	42.8	31.5
		MCWB	27.6	30.0	34.9	45.1	55.7	61.4	65.3	65.7	60.9	50.6	41.1	30.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.9	39.1	50.8	61.2	67.4	70.7	74.0	73.0	69.3	62.1	53.8	47.8
		MCDB	39.4	42.1	53.9	72.9	72.9	79.5	82.8	79.2	73.7	65.9	56.1	48.8
	2%	WB	33.4	35.0	44.1	55.9	63.2	68.5	71.7	70.1	67.0	58.8	50.2	38.4
		MCDB	34.3	36.8	48.0	63.3	69.3	76.8	80.1	75.3	71.3	61.8	51.8	39.2
	5%	WB	31.1	32.5	39.2	50.3	60.6	66.3	69.7	68.3	64.4	55.3	45.5	33.1
		MCDB	32.0	34.1	42.1	56.8	68.2	74.9	75.7	73.8	69.0	58.1	46.6	34.2
	10%	WB	27.7	29.6	35.1	46.0	57.9	63.9	67.6	66.6	61.6	51.6	41.1	30.6
		MCDB	28.9	31.8	37.9	52.5	64.9	70.0	73.9	72.0	65.7	54.6	43.1	31.5
Mean Daily Temperature Range	5% DB	MDBR	18.0	18.5	19.1	17.6	20.5	19.4	19.2	18.2	17.8	15.6	12.8	16.4
		MCDBR	19.0	16.1	20.9	27.6	26.9	23.7	22.5	20.9	21.3	21.2	19.7	18.4
		MCWBR	19.0	14.1	15.2	16.5	14.1	12.0	11.0	10.5	13.5	14.7	18.4	18.2
	5% WB	MCDBR	18.7	15.2	19.3	25.4	21.5	20.7	18.8	17.0	19.6	18.1	18.4	18.1
		MCWBR	18.9	14.2	15.6	16.8	12.7	11.9	10.7	9.5	13.8	14.7	18.3	18.1
Clear-Sky Solar Irradiance	taub	0.243	0.269	0.303	0.328	0.357	0.400	0.389	0.379	0.348	0.319	0.291	0.261	
	taud	2.354	2.263	2.269	2.424	2.413	2.286	2.323	2.356	2.438	2.532	2.515	2.363	
	Ebn at Noon	279	289	293	296	289	275	277	275	276	270	259	256	
	Edh at Noon	23	31	36	34	36	41	39	36	30	24	20	21	
All-Sky Solar Radiation	RadAvg	433	739	1169	1443	1626	1778	1774	1564	1219	710	424	312	
	RadStd	60	88	106	134	160	131	157	75	104	65	54	26	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page